James Motes

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EDUCATION/TRAINING

Postdoctoral Researcher, University of Illinois Urbana Champaign August 2023 - Present
Advisor: Dr. Nancy M. Amato
Ph.D. in Computer Science, University of Illinois Urbana Champaign August 2023
Ph.D Thesis: Multi-robot Task and Motion Planning in Hybrid State Spaces
Advisor: Dr. Nancy M. Amato
M.S. in Computer Science, Texas A&M University ` August 2019
M.S. Thesis: Interaction Templates for Multi-Robot Systems
Advisor: Dr. Nancy M. Amato
B.S. in Computer Engineering, Texas A&M University May 2018
Minor in Mathematics, Engineering Honors, Undergraduate Research Scholar

RESEARCH INTEREST

Multi-Robot Planning: How do we exploit growing general intelligence within the complexity of multi-robot problem spaces?

Human-Robot Co-planning:: How do we design cooperative planning systems in which humans, robots, and AI services jointly create and refine plans?

Hardware and Software Acceleration of Planning: How can we design planning architectures that fully exploit modern hardware and software accelerations?

Applications: Smart manufacturing and assembly; logistics and warehousing; agriculture; automated labs; healthcare and in-home assistive robotics

RESEARCH EXPERIENCE

Postdoctoral Researcher, University of Illinois Urbana Champaign August 2023 - Present

- Leads multiple graduate research teams (17 graduate students across lifetime of projects)
- Primary author for grant proposals (Submissions to NSF Foundational Research on Robotics, 6 industry white papers)
- Leads software development for the open source deployment of the Parasol Planning Library (PPL)
- Develops general framework for autonomous factories applied to manufacturing, biofabrication labs, and extraterrestrial resource collection
- Designs natural language interfaces for human preference specification for robot motion behaviors in collaboration with Natural Language Processing faculty
- Leverages computational geometry techniques and GPU serialization for accelerating core motion planning algorithms in dynamic environments
- Evaluates the tradeoff of multi-robot planning paradigms in different settings
- Develops routing design tools for Three-Dimensional Spatial Packaging of Interconnected Systems with Physical interactions (3D SPI2)

Graduate Researcher, University of Illinois Urbana Champaign August 2019 - August 2023

- Collaborated with Foxconn Interconnect Technologies and two other research labs at UIUC (Dr. Timothy Bretl, Dr. Katherine Driggs-Campbell) on a series of projects for a collaborative human-robot factory
- Developed multi-agent interaction template method enabling multi-agent task planning through a single motion planning query rather than a traditional task and motion planning system
- Designed framework for optimal multi-robot planning for large task sets
- Developed complex precedence and synchronization task constraint handling for multi-robot systems

Graduate Researcher, Texas A&M University May 2018 - August 2019

- Developed the Interaction Template Method for resuing expensive precomputed coordinated interactions (e.g., handoffs) in multi-robot task and motion planning
- Culminated in master's thesis and IEEE RA-L publication
- Mentored undergraduate researchers who contributed to the publication

Undergraduate Research Scholar, Texas A&M University September 2017 - May 2018

- Developed proactive multi-agent persistent task performance system under battery constraint
- Culminated in undergraduate thesis

PUBLICATIONS (12, Google Scholar citations = 309, h = 7 as of 11/15/2025)

Journal Articles (9)

- Lee, H., **Motes, J. D.**, Morales, M., & Amato, N. M., An Analysis of Constraint-Based Multi-Agent Pathfinding Algorithms. (2026). To appear in *IEEE Transactions on Robotics*.
- Qin, M., Solis, I., **Motes, J. D**., Morales, M., & Amato, N. M. (2025). K-ARC: Adaptive Robot Coordination for Multi-Robot Kinodynamic Planning. *IEEE Robotics and Automation Letters*. To be presented at the International Conference on Robotics and Automation (ICRA), Vienna, Austria, June, 2026.
- Solis, I., **Motes, J.**, Qin, M., Morales, M., & Amato, N. M. (2024). Adaptive robot coordination: A subproblem-based approach for hybrid multi-robot motion planning. *IEEE Robotics and Automation Letters*. Presented at ICRA@40, Rotterdam, Netherlands, September, 2024.
- McBeth, C., **Motes, J**., Uwacu, D., Morales, M., & Amato, N. M. (2023). Scalable Multi-robot Motion Planning for Congested Environments With Topological Guidance. *IEEE Robotics and Automation Letters*. Presented at the International Conference on Robotics and Automation (ICRA), Yokohama, Japan, May, 2024.
- **Motes, J**., Chen, T., Bretl, T., Aguirre, M. M., & Amato, N. M. (2023). Hypergraph-based multi-robot task and motion planning. *IEEE Transactions on Robotics*. Presented at the International Conference on Robotics and Automation (ICRA), Yokohama, Japan, May, 2024.
- Lee, H., **Motes, J.,** Morales, M., and Amato, N. M., 2021. Parallel Hierarchical Composition Conflict-Based Search for Optimal Multi-Agent Pathfinding. *IEEE Robotics and Automation Letters*, 6(4), pp. 7001-7008. Presented at the International Conference on Intelligent Robots and Systems (IROS), Prague, Czech Republic, 2021 (virtual).
- Solis, I., **Motes, J.**, Sandström, R., and Amato, N. M., 2021. Roadmap-Optimal Multi-Robot Motion Planning using Conflict-Based Search. *IEEE Robotics and Automation Letters*, 6(3), pp. 4608-4615. Presented at the International Conference on Robotics and Automation (ICRA), Xi'an, China, 2021 (virtual).
- **Motes, J.**, Sandström, R., Lee, H., Thomas, S., and Amato, N. M., April 2020. "Multi-Robot Task and Motion Planning With Subtask Dependencies," in *IEEE Robotics and Automation Letters*, 5(2), pp. 3338-3345. Presented at the International Conference on Robotics and Automation, Paris (ICRA), France, 2020 (virtual).
- **Motes, J.**, Sandström, R., Adams, W., Ogunyale, T., Thomas, S. and Amato, N.M., 2019. Interaction Templates for Multi-Robot Systems. *IEEE Robotics and Automation Letters*, 4(3), pp. 2926-2933. Presented at the International Conference on Intelligent Robots and Systems (IROS), Macao, China, 2019.

Refereed Conference Proceedings (2)

- Goddu, A., Brown, T., Paton, M., **Motes, J.**, & Chen, T. (2025). Modifying ABIT* for Tethered Rappelling Robot Motion Planning. *In Proceedings of the 21st International Conference on Automation Science and Engineering (CASE)* (pp. 722-728). IEEE, 2025.
- Ashur, S., Lusardi, M., Markowicz, M., **Motes, J**., Morales, M., Har-Peled, S., & Amato, N. M. (2024). SPITE: Simple Polyhedral Intersection Techniques for modified Environments. *In Proceedings of the Twelfth Workshop on the Algorithmic Foundations of Robotics,* 2024.

Workshop Papers (1)

- Chen, T., Huang, Z., **Motes, J.**, Geng, J., Ta, Q., Dinkel, H., Abdul-Rashid, H., Myers, J., Mun, Y., Lin, W., Huang, Y., Liu, S., Morales, M., Amato, N. M., Driggs-Campbell, K., Bretl, T., 2022. Insights from an Industrial Collaborative Assembly Project: Lessons in Research and Collaboration. *IEEE ICRA Workshop on Collaborative Robots and the Work of the Future,* Philadelphia, USA, 2022.

Under Review (6 as of 11/05/2025)

- Ngui, I., McBeth, C., Santos, A., He, G., Mimnaugh, K. J., **Motes, J. D.**, Soares, L., Morales, M. and Amato, N.M., ERUPT: An Open Toolkit for Interfacing with Robot Motion Planners in Extended Reality. *arXiv preprint arXiv:2510.02464*.
- Lee, S., **Motes, J. D.**, Ngui, I., Morales, M., & Amato, N. M., Lazy-DaSH: Lazy Approach for Hypergraph-based Multi-robot Task and Motion Planning. *arXiv preprint arXiv:2504.05552*.
- Lee, H., Serlin, Z., **Motes, J. D.**, Long, B., Morales, M., & Amato, N. M., PRISM: Complete Online Decentralized Multi-Agent Pathfinding with Rapid Information Sharing using Motion Constraints. *arXiv preprint arXiv:2505.08025*.
- Solis, I., **Motes, J. D.**, Qin, M., Morales, M., & Amato, N. M., Experience-based Subproblem Planning for Multi-Robot Motion Planning. *arXiv preprint arXiv:2411.08851*.

- McBeth, C., **Motes, J. D.**, Ngui, I., Morales, M., & Amato, N. M., Scalable Multi-Robot Motion Planning Using Guidance-Informed Hypergraphs. *arXiv preprint arXiv:2311.10176*.
- Attali, A., Ashur, S., Love, I. B., McBeth, C., **Motes, J. D.**, Uwacu, D., Morales, M. and Amato, N.M., A Framework for Guided Motion Planning. *arXiv preprint arXiv:2404.03133*.

In Preparation for 2025 (5 as of 11/05/2025)

- **Motes, J. D.**, Markowicz, M., Ashur, S., Arad, Y., Morales, M., & Amato, N. M., sLRS: semi-Lazy Rearrangement Solver - Leveraging Dynamic Roadmap Graphs in Task and Motion Planning.
- Arad, Y., Markowicz, M., Ashur, S., **Motes, J. D.**, Morales, M., & Amato, N. M., Serialized Red-Green-Gray: A Quick Heuristic Validation of Edges in Dynamic Roadmap Graphs.
- Markowicz, M., Arad, Y., Ashur, S., **Motes, J. D.**, Morales, M., & Amato, N. M., Greedy Lazy SPITE: Heuristic Validation and Lazy Motion Planning with Dynamic Roadmap Graphs.
- Markowicz, M., **Motes, J. D.**, Solis, I., Morales, M., & Amato, N. M., MR-SPITE: Leveraging Dynamic Roadmap Graphs in Multi-robot Motion Planning.
- Love, I. B., Lohan, D. J., **Motes, J. D.**, Dede, E. M., Morales, M., & Amato, N. M., A Hierarchical Approach for the Generation of Varied Cable Harness Designs.

FUNDING AUTHORSHIP

National Science Foundation (NSF) - Foundational Research in Robotics (FRR) Proposal September 2025
- Primary author on proposal for customizing robotic motion behaviors for non-technical users via natural language preference specification
- Collaboration with Natural Language Processing researchers at University of Illinois Urbana Champaign

Industry Proposals and Whitepapers April 2025 - Present
- Primary author on six proposals (secondary author on three more) covering human-robot interaction, multi-robot planning, environment design, and leveraging data in planning

TALKS

- **February 2024 - Invited Lab Seminar.** “Multi-Robot Task and Motion Planning in Hybrid State Spaces.” *Kavraki Lab, Department of Computer Science, Rice University*, Houston, TX
- **October 2019 - Invited Workshop Talk.** “Planning Motions and Tasks for Manipulators, Multi-Robot Systems and Biomolecules.” *Center for Autonomy Portfolio Discovery Workshop, UIUC*, Urbana, IL

TEACHING & MENTORING EXPERIENCE

Houston Robotics Club August 2023 - Present
- Mentors community college and high school students on AI and robotics projects
- Several now enrolled in STEM undergraduate programs or at internships with local robotics companies

Postdoctoral Researcher, University of Illinois Urbana Champaign (UIUC) August 2023 - Present
- Mentors 17 graduate students (5 graduated), some carried over from Ph.D. Graduate Researcher
- Lead to 5 published manuscripts, 7 under review, and 5 in preparation for submission in 2025 (15 total)

Lead Instructor, AI4All August 2021 - December 2023
- Instructor for Discover AI introductory artificial intelligence courses at the University of Texas at El Paso and New Mexico State University (Spring 2022), Texas A&M University (Fall 2022, Spring 2023, Fall 2023) and Worcester Polytechnic Institute (Fall 2022 and Spring 2023)
- Developed course curriculum for Discover AI introductory artificial intelligence course (Summer 2022)
- Led and mentored team of instructors at other universities
- Teaching assistant for Discover AI introductory artificial intelligence course at the University of Illinois Urbana Champaign (Fall 2021)

Graduate Researcher, UIUC and Texas A&M University May 2018 - August 2023
- Mentored graduate students (14) within my research group
- Mentored numerous undergraduate students, both local university students and DREU program students
- Lead to 7 published manuscripts
- Taught motion planning course for new lab members (2019-2021)

Teaching Assistant, University of Illinois Urbana Champaign January 2021 - May 2021

- Computer Science PhD seminar course
- Organized guest speakers, managed grading and student assignments

START UP AND INDUSTRY EXPERIENCE

Optigon Inc, Principal Software Engineer May 2024 - Present

- Leading software development for automated metrology system for perovskite solar cell technologies
- Advising on machine learning-based analytical models of solar cell behaviors

Normandy Automation, Founder May 2023 - August 2025

- Developed autonomous robotic manufacturing solutions
- Leveraged AI and multi-robot systems for just-in-time manufacturing with minimal human reprogramming

QuickReports, Founder May 2023 - August 2025

- Developed software for healthcare management service organization business development teams
- Provided autonomously generated competition and demographics reports for brick and mortar healthcare practices

Hewlett Packard Enterprise - Design Verification Engineer Intern May - August 2017

- Worked on Gen-Z project within the Silicon Design Lab
- VLSI Design Verification utilizing Universal Verification Methodology (UVM)

HONORS

Engineering Graduate Merit Fellowship August 2018 - August 2019
Presidential Endowed Scholarship August 2014 - May 2018
Distinguished Student Award - Dwight Look College of Engineering May 2017
Industrial Affiliates Program Scholarship August 2014 - May 2018
Dell Merit Scholarship August 2017 - May 2018
Salutatorian - Rowlett High School (2/591) June 2014

SERVICE AND PROFESSIONAL ACTIVITIES

Program Committee Member

- WAFR 2026
- WAFR 2024

Reviewer/Referee August 2018 - Present

- IEEE Robotics and Automation Letters
- IEEE Transactions on Robotics
- IEEE International Conference on Robotics and Automation
- IEEE International Conference on Intelligent Robotics and Systems
- IEEE International Symposium on Multi-robot and Multi-agent Systems
- International Workshop on Algorithmic Foundations of Robotics
- International Journal of Robotics Research

Member of IEEE Robotics and Automation Society May 2019 - Present